Question:
A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?
Answer Choices:
A. Decreased prolactin, decreased FSH, decreased LH
B. Decreased prolactin, increased FSH, increased LH
C. Increased prolactin, decreased FSH, increased LH
D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>
Finding Reasoning Paths:
1. **Galactorrhea + Abnormal Breast Secretions + Missing Menstrual Periods**: This combination suggests a hypersecretory breast disorder, such as a prolactin-secreting pituitary tumor or other causes like primary hypothyroidism, exogenous estrogen intake, or medications like risperidone or tramadol.
2. **Prolactinoma (Prolactin-secreting Tumor)**: A prolactinoma is the most common cause of hyperprolactinemia and is associated with galactorrhea, amenorrhea, and infertility. MRI of the brain may show a pituitary mass.
3. **Hyperprolactinemia**: Other causes of hyperprolactinemia include medications (e.g., dopamine-blocking agents like risperidone, antipsychotics, or tramadol), thyroid dysfunction, and hypothyroidism.
4. **Primary Hypothyroidism**: This can also lead to hyperprolactinemia and is a common endocrine disorder associated with galactorrhea and amenorrhea.
5. **Psychological Factors**: Stress, depression, and other emotional states can lead to hyperprolactinemia, though this is less likely given the persistent nature of the symptoms.

Reasoning Process:
1. **Key Symptoms**: The patient presents with galactorrhea (milky-white nipple discharge), amenorrhea (missed menstrual periods), and hyperprolactinemia. These symptoms are highly suggestive of a hypersecretory breast disorder, particularly a prolactin-secreting pituitary tumor (prolactinoma).
2. **Prolactinoma**: A prolactinoma is a benign tumor of the anterior pituitary that secretes excess prolactin. This can lead to a constellation of symptoms, including:
- Galactorrhea (due to increased prolactin stimulating milk production).
- Amenorrhea and infertility (due to negative feedback on the hypothalamic-pituitary-ovarian axis).
- Abnormal breast secretions (e.g., creamy or watery discharge).
- MRI of the brain may show a pituitary mass.

3. **Hyperprolactinemia**: The patientâĢĻs symptoms of hyperprolactinemia (galactorrhea, amenorrhea) align with a prolactinoma. However, other causes of hyperprolactinemia should be considered, such as medications (e.g., risperidone, antipsychotics, tramadol), exogenous estrogen, or primary hypothyroidism. These would also explain the galactorrhea and amenorrhea, but MRI would typically show no pituitary mass in these cases.

4. **MRI of the Brain**: If the MRI shows a pituitary mass, a prolactinoma is the most likely diagnosis. Prolactin levels are typically elevated in patients with prolactinomas, as the tumor produces excess prolactin.

5. **Exclusion of Other Causes**:
- **Hypothyroidism**: While hypothyroidism can lead to hyperprolactinemia, it is less likely to explain the galactorrhea and amenorrhea alone. Additionally, hypothyroidism typically presents with thyroid-related symptoms (e.g., fatigue, cold intolerance).
- **Medications**: Medications like risperidone or antipsychotics can cause hyperprolactinemia and galactorrhea, but they typically do not cause amenorrhea.
- **Psychological Factors**: Stress and depression can lead to hyperprolactinemia, but they are not associated with galactorrhea or amenorrhea.

Conclusion:
Based on the clinical presentation of galactorrhea, amenorrhea, and hyperprolactinemia